

Design and Fabrication of Square Loop Antenna (433 MHz)

1. Introduction: A square loop antenna is a closed-loop conductor that radiates electromagnetic waves when current flows through it. It behaves as a magnetic dipole.

2. Working Principle: Alternating current produces a time-varying magnetic field which generates electromagnetic radiation. Maximum radiation is perpendicular to the loop.

3. Frequency Selection: 433 MHz is chosen for easy testing with RF modules.

4. Design Calculation: Wavelength $\lambda = c/f \approx 0.69$ m. Each side ≈ 17 cm.

5. Components Required: Copper wire, coaxial cable, SMA connector, RF transmitter/receiver module, soldering tools.

6. Fabrication Steps: Cut wire, form square loop, create feed gap, connect coaxial cable, mount structure.

7. Signal Transmission: RF transmitter sends signal → antenna radiates → receiving antenna captures → RF receiver processes signal.

8. Simulation Software: ANSYS HFSS and CST Studio Suite for radiation pattern and gain analysis.

9. Applications: Wireless communication, RF systems, signal detection.

10. Conclusion: A simple and effective antenna suitable for educational and practical use.